Lectures 23-24

Math 4741: Mathematical Analysis

Chapter 9: Markov Decision Processes

Reference

Markov Processes for Stochastic Modeling – Oliver Ibe (Chapter 13)

Markov Decision Process (MDP)

- An MDP Models an environment where an agent takes a sequence of decisions
- The agent observes the system state at specific points in time (decision epochs) and gathers information to take the decision
- Each action is associated with a cost (positive or negative)
    - also known as reward
- The ultimate objective is to take decisions so that the reward is maximized

MDP Components:

States (S): a set of finite states representing the system conditions

Actions (A): a finite set of actions available to the agent

Probability Model (Transition Model): A function that defines the probability of transition from one state to another given a specific action [f(S, a, S')]

Reward (R): a function that assigns a numerical reward to each state (and possibly each action) to quantify the immediate benefit [f(s,a)]

Discount factor (β): A value between 0 and 1 representing the importance of the future rewards as compared to the immediate reward

Discrete vs Continuous Time

If T denotes the set of decision epochs, then an MDP can be

- Discrete time, if decisions are made at each decision epoch
- Continuous time, if decisions are made continuously or when some events occur

For discrete time MDP,

- T can be finite, $T = 1, 2, \ldots, N$ and $N < \infty$, the decision process is called a finite horizon (or finite stage) decision process
- T can be infinite, $T = 1, 2, \ldots,$ the decision process is called an infinite horizon (infinite state) decision process

Policy:

A rule for making decisions at each decision epoch is called a policy

And, is the strategy that specifies the action to take in each state that maximizes the expected sum of rewards.

$D = (d_1, d_2, \dots, d_{N-1})$ denotes the policy, where d_t is the action to be taken at decision epoch $t \in T$

A policy can be stationary and non-stationary

A stationary policy takes the same decision a_i, when the system is in state i

Applications of MDPs:

- Robotics and automated control systems
- Operations research for decision-making processes
- Game theory and artificial intelligence for strategic planning
- Economics and finance for optimizing investment strategies

Solving MDPs:

Common methods to solve MDPs include:

- **Value Iteration**: An iterative algorithm that updates the value of each state until convergence.
- **Policy Iteration**: Alternates between evaluating a policy and improving it until the best policy is found.
- **Linear Programming**: A systematic method that breaks down the problem into simpler subproblems.

Example of DP problem

Consider a situation where we are given X units of a resource to be allocated to N activities. For example, we might want $\$X$ to be used to fund N projects. Suppose we are also given a table that lists the return $r_i(x)$ to be realized from allocating x units of the resource to activity i, where $i = 1, 2, \ldots, N$ and $x = 0, 1, \ldots, X$. Then the problem we are confronted with becomes the following:

$$\text{Maximize} \sum_{i=1}^{N} r_i(x_i)$$

$$\text{Subject to} \sum_{i=1}^{N} x_i = X$$

$$0 \leq x_i \leq X \quad i = 1, 2, \ldots, N$$

the quantity left. We define the optimal value function $v_k(x)$ as the maximum return obtained from activities k through N, given that x units of the resource remain to be allocated. From the principle of optimality, we obtain the following recurrence relation

$$v_k(x) = \max[r_k(x_k) + v_{k+1}(x - x_k)] \quad x_k = 0, 1, \ldots, x$$

where x_k is the allocation to activity k and $x = 0, 1, \ldots, X$. The boundary condition is

$$v_N(x) = r_N(x)$$

and the solution to the problem is $v_1(x)$.

Example 13.1

For a numerical example, we consider $X = 6$ and $N = 3$. Table 13.1 gives the values of the returns $r_i(x)$.

Table 13.1 Data for the Numerical Example

x_1	$r_1(x_1)$	x_2	$r_2(x_2)$	x_3	$r_3(x_3)$
0	0	0	0	0	0
1	3	1	2	1	1
2	6	2	4	2	3
3	9	3	6	3	5
4	12	4	9	4	8
5	16	5	11	2	12
6	16	6	13	6	13

x_1	$r_1(x_1)$	x_2	$r_2(x_2)$	x_3	$r_3(x_3)$
0	0	0	0	0	0
1	3	1	2	1	1
2	6	2	4	2	3
3	9	3	6	3	5
4	12	4	9	4	8
5	16	5	11	2	12
6	16	6	13	6	13

$v_n(x)$ = Maximum return from activities k through N given x units of resources allocated

$$v_k(x) = \min_{0 \le x_k \le x} [r_k(x_k) + v_{n-1}(x - x_k)] \quad 0 \le x \le X$$

$$v_N(x) = r_N(x)$$

$$v_3(0) = 0$$

$$v_3(1) = 1$$

$$v_3(2) = 3$$

$$v_3(3) = 5$$

$$v_3(4) = 8$$

$$v_3(5) = 12$$

$$v_3(6) = 13$$

$$v_2(0) = 0, \quad m_2(0) = 0$$

$$v_2(1) = \max[r_2(x_k) + v_3(1 - x_k)] = \max[r_2(0) + v_3(1), r_2(1) + v_3(0)]$$
$$= \max[0 + 1, 2 + 0] = 2, m_2(1) = 1$$

$$v_2(2) = \max[r_2(0) + v_3(2), r_2(1) + v_3(1), r_2(2) + v_3(0)]$$
$$= \max[0 + 3, 2 + 1, 4 + 0] = 4, m_2(2) = 2$$

$$v_2(3) = \max[r_2(0) + v_3(3), r_2(1) + v_3(2), r_2(2) + v_3(1), r_2(3) + v_3(0)]$$
$$= \max[0 + 5, 2 + 3, 4 + 1, 6 + 0] = 6, m_2(3) = 3$$

$$v_2(4) = \max[r_2(0) + v_3(4), r_2(1) + v_3(3), r_2(2) + v_3(2), r_2(3) + v_3(1), r_2(4) + v_3(0)]$$
$$= \max[0 + 8, 2 + 5, 4 + 7, 6 + 1, 9 + 0] = 9, m_2(4) = 4$$

$$v_2(5) = \max[r_2(0) + v_3(5), r_2(1) + v_3(4), r_2(2) + v_3(3), r_2(3) + v_3(2),$$
$$r_2(4) + v_3(1) + r_2(5) + v_3(0)]$$
$$= \max[0 + 12, 2 + 8, 4 + 5, 6 + 3, 9 + 1, 11 + 0] = 12, m_2(5) = 0$$

$$v_2(6) = \max[r_2(0) + v_3(6), r_2(1) + v_3(5), r_2(2) + v_3(4), r_2(3) + v_3(3),$$
$$r_2(4) + v_3(2) + r_2(5) + v_3(1), r_2(6) + v_3(0)]$$
$$= \max[0 + 13, 2 + 12, 4 + 8, 6 + 5, 9 + 3, 11 + 1, \quad 13 + 0] = 14, m_2(6) = 1$$

$$
\begin{aligned}
v_1(6) &= \max[r_1(0) + v_2(6), r_1(1) + v_2(5), r_1(2) + v_2(4), r_1(3) + v_2(3), \\
&\qquad r_1(4) + v_2(2) + r_1(5) + v_2(1), r_1(6) + v_2(0)] \\
&= \max[0 + 14, 2 + 12, 6 + 9, 9 + 6, 12 + 4, 16 + 2, 16 + 0] = 18, m_1(6) = 5
\end{aligned}
$$

Markov Reward Process (MRP)

MRP is an extension of the basic Markov process $\{X_n, n = 1, 2. \dots\}$ that associates each state with a reward

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \cdots & r_{NN} \end{bmatrix}$$

$\{X_n, R\}$ is a discrete-time MRP

$$
\begin{aligned}
v_n(i) &= E[\Re_n + \Re_{n-1} + \cdots + \Re_1 + \Re_0 | s_n = i] \\
&= E[\{\Re_n | s_n = i\} + \{\Re_{n-1} + \cdots + \Re_1 + \Re_0 | s_n = i\}] \\
&= E[\Re_n | s_n = i] + E[\Re_{n-1} + \cdots + \Re_1 + \Re_0 | s_n = i] \\
&= \sum_{j=1}^{N} p_{ij} r_{ij} + \sum_{j=1}^{N} p_{ij} v_{n-1}(j)
\end{aligned}
$$

- The first sum denotes the expected immediate reward by making a transition from i to any state
  - # of remaining transition is $n - 1$
- The 2nd sum represents the expected total reward in these $n - 1$ transitions, given that the process is currently in sate j, $v_{n-1}(j)$

$$q_i = \sum_{j=1}^{N} p_{ij} r_{ij}$$

$$v_n(i) = q_i + \sum_{j=1}^{N} p_{ij} v_{n-1}(j)$$

$$v_n = q + P v_{n-1}$$

MDP basics

An extension of both MRP and DP

- S is a finite set of N states, that is, $S = \{1, 2, \ldots, N\}$. In practice, the state of a system is a set of parameters that can be used to describe the system. For example, the state of a robot can be the coordinates of the robot.
- A is a finite set of K actions that can be taken at any state; that is, $A = \{a_1, a_2, \ldots, a_K\}$. In the case of a robot that can move in discrete steps, for example, an action can be a statement like "go east" or "go west."
- R is the reward matrix, which can vary with the action taken. Thus, for action $a \in A$, we denote the reward associated with a transition from state i to state j when action a is taken by $r_{ij}(a)$.
- P is the transition probability matrix, which can be different for each action. Thus, for action $a \in A$, we denote the probability that the process moves from state i to state j when action a is taken by $p_{ij}(a)$.

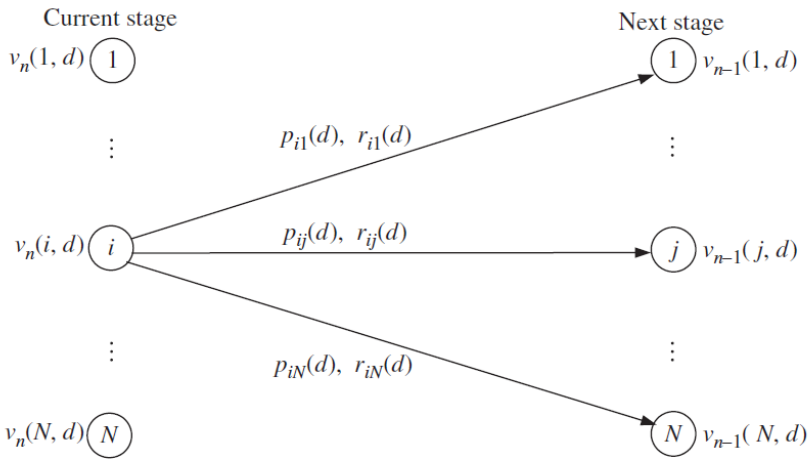
For such a system, we can see that

$$P[S_{n+1} = j | S_0, a_0, S_1, a_1, \ldots, S_n = i, a_n = a] = P[S_{n+1} = j | S_n = i, a_n = a] = p_{ij}(a)$$

Thus, the transition probabilities and reward functions are functions only of the last state and the subsequent action. Any homogeneous Markov chain $\{S_n\}$ whose transition probabilities are $p_{ij}(a)$ is called an MDP, where

$$\sum_j p_{ij}(a) = 1 \quad \text{for all } i \in S \text{ and all } a \in A$$

$$v_n(i,d) = \sum_{j=1}^{N} p_{ij}(d) r_{ij}(d) + \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j,d) = q_i(d) + \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j,d)$$



$$q_i(d) = \sum_{j=1}^{N} p_{ij}(d) r_{ij}(d)$$

$$v_n(i) = \max_d \left\{ q_i(d) + \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j) \right\} \quad i = 1, 2, \ldots, N$$

MDP with Discounting

$$
\begin{aligned}
v_n(i,d,\beta) &= E[\mathfrak{R}_n + \beta\mathfrak{R}_{n-1} + \beta^2\mathfrak{R}_{n-2} + \cdots + \beta^{n-1}\mathfrak{R}_1 + \beta^n\mathfrak{R}_0 | s_n = i] \\
&= E[\{\mathfrak{R}_n | s_n = i\} + \beta\{\mathfrak{R}_{n-1} + \beta\mathfrak{R}_{n-2} + \cdots + \beta^{n-2}\mathfrak{R}_1 + \beta^{n-1}\mathfrak{R}_0\} | s_n = i] \\
&= E[\{\mathfrak{R}_n | s_n = i\}] + \beta E[\{\mathfrak{R}_{n-1} + \beta\mathfrak{R}_{n-2} + \cdots + \beta^{n-2}\mathfrak{R}_1 + \beta^{n-1}\mathfrak{R}_0\} | s_n = i] \\
&= \sum_{j=1}^{N} p_{ij}(d) r_{ij}(d) + \beta \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j,d,\beta) \\
&= q_i(d) + \beta \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j,d,\beta)
\end{aligned}
$$

MDP solution Methods:

- Value iteration method: finite horizon problems
- Policy iteration method: infinite horizon problems
- Linear programming: infinite horizon problems

Value iteration method

$$v_n(i, d, \beta) = q_i(d) + \beta \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j, d, \beta)$$

$$v_n(i, \beta) = \max_d \left\{ q_i(d) + \sum_{j=1}^{N} p_{ij}(d) v_{n-1}(j, \beta) \right\} \quad i = 1, 2, \ldots, N$$

It starts by choosing a set of values of $v_0(i, d, \beta)$ as follows:

$$v_0(1, d, \beta) = v_0(2, d, \beta) = \cdots = v_0(N, d, \beta) = 0$$

Thus, we can obtain $v_1(i, \beta)$ as follows:

$$v_1(i, \beta) = \max_d \{q_i(d)\} \quad i = 1, 2, \ldots, N$$

This gives the expected total return in stage 1, given that the process is in state i at that stage when the optimal policy is used. Using this set of $v_1(i, \beta)$, we obtain $v_2(i, \beta)$, the expected total return in stage 2, as follows:

$$v_2(i, \beta) = \max_d \left\{ q_i(d) + \sum_{j=1}^{N} p_{ij}(d) v_1(j, \beta) \right\} \quad i = 1, 2, \ldots, N$$

$$v_T \longleftarrow v_{T-1} \longleftarrow \cdots \longleftarrow v_2 \longleftarrow v_1 \longleftarrow v_0$$

Figure 13.2 Recursive procedure for the expected return.

Example: An Equipment that is inspected daily and can be in one of three states – Good, Acceptable and Bad

a. Do nothing, and thus follow the transition probabilities defined above.
b. Overhaul, which is equally likely to bring it to the Good condition or the Acceptable condition; the cost of an overhaul is $500.
c. Replace the equipment, which automatically brings it to the Good condition; the cost of a new equipment is $2000.

Assume that when the equipment is operating in Good condition, the company makes $1000. When it is operating in Acceptable condition, the company makes $500, and when it is operating in Bad condition, the company loses $500. We consider the following policies:

a. Replace the equipment only when it is in Bad condition and nothing in other states.
b. Replace the equipment when it is in Bad condition and overhaul when it is in Acceptable condition.
c. Replace the equipment when it is in Acceptable condition and when it is in Bad condition.

Determine the optimal operating cost of the equipment when $T = 4$.

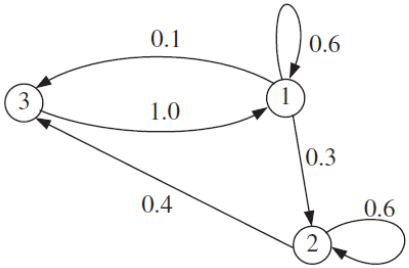
Solution

Let the states be defined as follows:

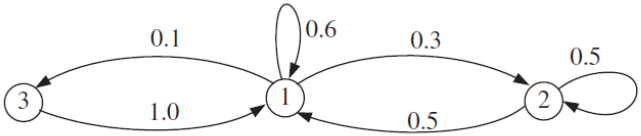
$1 \equiv$ Good

$2 \equiv$ Acceptable

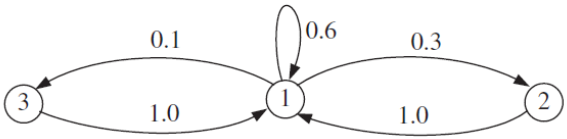
$3 \equiv$ Bad



(A)



(B)



(C)

$$P(1) = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.0 & 0.6 & 0.4 \\ 1.0 & 0.0 & 0.0 \end{bmatrix} \quad R(1) = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & -1 \\ -2 & 0 & 0 \end{bmatrix} \quad Q(1) = \begin{bmatrix} 1.4 \\ 0.2 \\ -2 \end{bmatrix}$$

$$P(2) = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.5 & 0.5 & 0.0 \\ 1.0 & 0.0 & 0.0 \end{bmatrix} \quad R(2) = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \quad Q(2) = \begin{bmatrix} 1.4 \\ 0.5 \\ -2 \end{bmatrix}$$

$$P(3) = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 1.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 \end{bmatrix} \quad R(3) = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \quad Q(3) = \begin{bmatrix} 1.4 \\ 1 \\ -2 \end{bmatrix}$$

Stage 1

	$v_1(i, d) = q_i(d)$			Optimal Solution	
i	$d = 1$	$d = 2$	$d = 3$	$v_1(i)$	d^*
1	1.4	1.4	1.4	1.4	1,2,3
2	0.2	0.5	1	1	3
3	−2	−2	−2	−2	1,2,3

Stage 2

	$v_2(i, d) = q_i(d) + p_{i1}(d)v_1(1) + p_{i2}(d)v_1(2) + p_{i3}(d)v_1(3)$			Optimal Solution	
i	$d = 1$	$d = 2$	$d = 3$	$v_2(i)$	d^*
1	2.34	2.34	2.34	2.34	1,2,3
2	0	1.7	2.4	2.4	3
3	−0.6	−0.6	−0.6	−0.6	1,2,3

Stage 3

	$v_3(i, d) = q_i(d) + p_{i1}(d)v_2(1) + p_{i2}(d)v_2(2) + p_{i3}(d)v_2(3)$			Optimal Solution	
i	$d = 1$	$d = 2$	$d = 3$	$v_2(i)$	d^*
1	3.464	3.464	3.464	3.464	1,2,3
2	0.80	2.87	3.34	3.34	3
3	0.34	0.34	0.34	0.34	1,2,3

Stage 4

	$v_4(i, d) = q_i(d) + p_{i1}(d)v_3(1) + p_{i2}(d)v_3(2) + p_{i3}(d)v_3(3)$			Optimal Solution	
i	$d = 1$	$d = 2$	$d = 3$	$v_2(i)$	d^*
1	4.5144	4.5144	4.5144	4.5144	1,2,3
2	2.340	3.902	4.464	4.464	3
3	1.464	1.464	1.464	1.464	1,2,3

The optimal solution shows that the user should do nothing when the equipment is Good in any year, replace the equipment if it is Acceptable in any year, and replace the equipment if it is Bad in any year. The total expected return after 4 years is $v_4(1) = 4.5144$ units if the equipment is Good in the first year, $v_4(2) = 4.464$ units if it is Acceptable in the first year, and $v_4(3) = 1.464$ units if it is Bad in the first year.